INCH-POUND

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PERFORMANCE SPECIFICATION

CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, MOLDED, CONFORMAL COATED AND METAL CASED WITH PLASTIC END-FILL, NONHERMETICALLY SEALED, GENERAL SPECIFICATION FOR

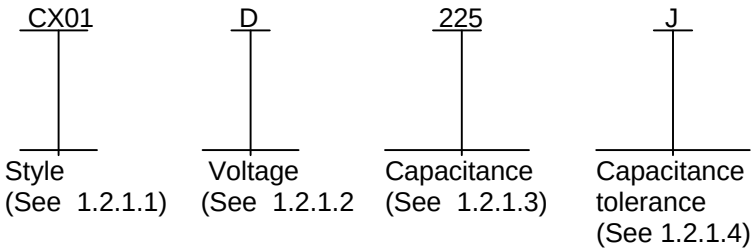
This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for fixed, electrolytic (solid electrolyte), tantalum, molded, conformal coated and metal cased with plastic end-fill nonhermetically sealed capacitors, primarily intended for filter, bypass, coupling, and other applications where the alternating current (ac) component is small compared to the direct current (dc) rated voltage. These capacitors are intended to be used only where supplemental moisture protection is provided or for noncritical applications where hermetic moisture protection is not required.

1.2 Classification. Capacitors covered by this specification are classified by the style, as specified (see 3.1).

1.2.1 Part or Identifying Number (PIN). The PIN will be in the following form and as specified (see 3.1 and 6.2).



Beneficial comments (recommendations additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Communications - Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703-5023, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5910

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TABLE I. Voltage.

Symbol	Rated (85° C) Volts, dc	Surge (85° C) Volts, dc
A	2	2.6
B	3	4.0
C	4	5.0
D	6	8.0
F	10	13.0
H	15	20.0
J	20	26.0
K	25	32.0
M	35	46.0
N	50	65.0

1.2.1.1 Style. The style is identified by the two-letter symbol CX, followed by a two-digit number; the letters identify fixed, electrolytic (solid electrolyte) tantalum, molded, conformal coated and metal cased with plastic end-fill, nonhermetically sealed capacitors, and the number identifies the design of the capacitor.

1.2.1.1 Voltage. The voltage (rated and surge) is identified by a single letter as shown in table I.

1.2.1.3 Capacitance. The nominal capacitance value, expressed in picofarads (pF) is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow.

1.2.1.4 Capacitance tolerance. The capacitance tolerance is identified by a single letter as shown in table II.

TABLE II. Capacitance tolerance.

Symbol	Capacitance tolerance
	Percent (±)
K	10
M	20

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in sections 3 and 4 of this specification, whether or not they are listed.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

(See supplement 1 for list of associated specification sheets.)

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (NPM-DODSSP), 700 Robbins Avenue, Philadelphia PA 19111-5094).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/J-STD-006 - Requirements for Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications.
ANSI/NCSL Z540-1 - General Requirements for Calibration Laboratories and Measuring and Test Equipment.

(Application for copies should be addressed to the American National Standards Institute (ANSI), 11 West 42nd Street, New York, NY 10036.)

INTERNATIONAL ORGANIZATION FOR STANDARDS (ISO)

ISO 10012-1 Quality Assurance Requirements for Measuring Equipment, Part 1: Meteorological Confirmation System for Measuring Equipment.

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets) the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specifications. The individual item requirements shall be as specified herein and in accordance with the applicable associated specifications. In the event of any conflict between requirements of this specification and the associated specifications, the latter shall govern (see 6.2).

3.2 Qualification. Capacitors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list before contract award (see 4.4 and 6.3).

3.3 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product. The use of paper products is not acceptable.

3.3 1 Solder and soldering flux. Solder and soldering flux shall be of such quality as to enable the capacitors to meet all of the requirements of this specification.

3.4 Interface and physical dimension requirements. Capacitors shall meet the interface requirements and physical dimensions specified (see 3.1).

3.4.1 Body structure. The body structure shall be of encapsulated form (see 3.1).

3.4.2 Terminals. Terminals shall be of a solid conductor, of the dimensions specified (see 3.1), and shall be suitably treated to facilitate soldering.

3.4.2.1 Solder dip (retinning). Only the manufacture may solder dip/retin the lead of the capacitors supplied to this specification, provided the solder dip process has been approved by the qualifying activity.

3.4.2.1.1 Qualifying activity approval. Approval of the solder dip process shall be based on one of the following options:

a. When the original lead finish qualified was hot solder dip lead finish 52 in accordance with MIL-STD-1276 (NOTE: The 200 microinch maximum thickness is not applicable). The manufacturer shall use the same solder dip process for retinning as was used in the original manufacture of the capacitor.

b. When the lead originally qualified was not solder dip lead finish 52 of MIL-STD-1276 as prescribed in 3.4.2.1.1a, approval for the process to be used for solder dip shall be based on the following procedure:

(1) Thirty samples of any capacitance value for each style and lead finish shall be subjected to the manufacturers solder dip process. The capacitors shall be subjected to all prescreen electrical tests, with no defects allowed.

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(2) Ten of the 30 samples shall be subjected to the solderability test, with no defects allowed.

(3) The remaining 20 samples shall be subjected to the resistance to soldering heat test, followed by the moisture resistance test, with no defects allowed.

3.4.2.1.2 Solder dip/retinning options. The capacitor manufacturer may solder dip/retin after the prescreen tests or as a corrective action. After solder dip/retinning, the following inspection and tests shall be performed on 100 percent of the lot:

DC Leakage: Shall not exceed the requirement specified in 3.7.

Capacitance: Shall be within the tolerance specified (see 3.1).

Dissipation factor: Shall not exceed the requirement specified in 3.9.

3.4.2.2 Tin plated finishes. Tin plating is prohibited as a final finish or as an undercoat. Tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is three percent (see 6.3).

3.5 Voltage aging. When tested as specified in 4.8.2, capacitors shall withstand the application of voltage at high temperature without visible mechanical damage.

3.6 Construction analysis (metal cased units only). When capacitors are inspected as specified in 4.8.3, construction analysis shall disclose no evidence of voids of the plastic end-fill exceeding 20 percent of the total case volume.

3.7 DC leakage. When measured as specified in 4.8.4, the dc leakage shall not exceed the applicable value specified (see 3.1).

3.8 Capacitance. When measured as specified in 4.8.5, the capacitance shall be within the applicable tolerance specified (see 3.1).

3.9 Dissipation factor. When measured as specified in 4.8.6, the dissipation factor shall not exceed the value specified (see 3.1).

3.10 Shock (specified pulse). When capacitors are tested as specified in 4.8.7, there shall be no intermittent contacts of 0.5 millisecond (ms) or greater duration, or arcing or other indication of breakdown, nor shall there be any open-or short-circuiting or evidence of mechanical damage.

3.11 Vibration, high frequency. When capacitors are tested as specified in 4.8.8, there shall be no intermittent contacts of 0.5 millisecond (ms) or greater duration, or arcing or other indication of breakdown, nor shall there be any open- or short-circuiting or evidence of mechanical damage.

3.12 Thermal shock. When tested as specified in 4.8.9, capacitors shall meet the following requirements:

DC leakage: Shall not exceed the requirement specified in 3.7.

Capacitance: Shall not change more than ±10 percent from the initial measured value (see 3.8).

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Dissipation factor: Shall not exceed the requirement specified in 3.9.

Visual examination: There shall be no evidence of harmful corrosion, mechanical damage, and obliteration of marking wherever applicable.

3.13 Resistance to soldering heat. When capacitors are tested as specified in 4.8.10, there shall be no evidence of mechanical damage.

3.14 Terminal strength. When capacitors are tested as specified in 4.8.11, there shall be no loosening of the terminals or permanent damage to the terminals.

3.15 Moisture resistance. When tested as specified in 4.8.12, capacitors shall meet the following requirements:

DC leakage: Shall not exceed 300 percent of the requirement specified in 3.7.

Capacitance: Shall not change more than ±15 percent from the initial measured value (see 3.1 and 3.8).

Dissipation factor: Shall not exceed 150 percent of the requirement specified in 3.9.

Visual examination: There shall be no evidence of harmful corrosion, mechanical damage, or obliteration of marking wherever applicable.

3.16 Stability at low and high temperatures. When tested as specified in 4.8.13, capacitors shall meet the following requirements:

Step 1 (+25° C):

DC leakage: Shall not exceed the applicable value specified (see 3.1).

Capacitance: Shall be within tolerance of the nominal value specified (see 3.1).

Dissipation factor: Shall not exceed the applicable value specified (see 3.1).

Step 2 (-55° C):

Capacitance: Shall not change more than the applicable value specified (see 3.1) from the step 1 measured value.

Dissipation factor: Shall not exceed the applicable value specified (see 3.1).

Step 3 (+25° C):

DC leakage: Shall not exceed the applicable value specified (see 3.1).

Capacitance: Shall not change more than 10 percent from the step 1 measured value.

Dissipation factor: Shall not exceed the requirement specified in 3.9.

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Step 4 (+85° C):

DC leakage: Shall not exceed the applicable value specified (see 3.1)
Capacitance: Shall not change more than the applicable value specified (see 3:1) from the step 1 measured value.

Dissipation factor: Shall not exceed the requirement specified in 3.9.

Step 5 (+25°C):

DC leakage: Shall not exceed the applicable value specified (see 3.1).

Capacitance: Shall not change more than the applicable value specified (see 3.1) from the step 1 measured value.

Dissipation factor: Shall not exceed the applicable value specified (see 3.1).

3.17 Surge voltage. When tested as specified in 4.8.14, capacitors shall meet the following requirements.

DC leakage: Shall not exceed the requirement specified in 3.7.

Capacitance: Shall not change more than the applicable value specified (see 3.1) from the initial measured value.

Dissipation factor: Shall not exceed the requirement specified in 3.9.

3.18 Life. When tested as specified in 4.8.15 there shall be no evidence of harmful corrosion, mechanical damage, or obliteration of marking (if applicable) and capacitors shall meet the following requirements:

DC leakage (at 25°C): Shall not exceed the applicable value specified (see 3.1).

Capacitance (at 25°C): Shall not change more than the applicable value specified (see 3.1) from the initial measured value.

Dissipation factor (at 25°C): Shall not exceed the applicable value, specified (see 3.1).

3.19 Solderability. When capacitors are tested as specified in 4.8.16, the criteria for evaluation shall be as specified in method 208 of MIL-STD-202.

3.20 Resistance to solvents (see 4.8.17). When tested as specified in 4.8.17, marking shall remain legible and shall not smear, and capacitors shall meet the following requirements:

DC leakage: Shall not exceed the requirement specified in 3.7

Capacitance: Shall not change more than ±2 percent from the initial measured value.

Dissipation factor: Shall not exceed the requirement specified in 3.9.

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3.21 Marking. Capacitors shall be marked in accordance with method I of MIL-STD-1285. Polarity for polarized units shall be indicated by a plus (+) symbol near the positive terminal. At the option of the manufacturer, a contrasting color may also be used to indicate polarity. Paper labels shall not be used. Marking shall remain legible after all tests. Capacitor marking shall include the specified information, in the sequence shown in the following example.

105K - Coded capacitance (in picofarads); capacitance tolerance.

+50T - Polarity 1/, voltage, manufacturers' trade mark.

3.22 Workmanship. Capacitors shall be processed in such a manner that when examined under 3X magnification, they shall be uniform in quality and shall be free from pits, cracks, rough edges, and other defects that will affect life, serviceability, or appearance.

3.22.1 Soldering. All excess flux or solder shall be removed. Electrical connections shall be electrically continuous after soldering.

3.23 Recycling and waste prevention. Recovered materials or environmentally preferable materials shall be used whenever possible without jeopardizing the intended use of the item.

4. VERIFICATION

4.1 Test equipment and inspection facilities. The supplier shall establish and maintain a calibration system in accordance with ANSI/NCSL Z540-1, ISO-10012-1, or equivalent system.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Retention of qualification (see 4.5).

c. Conformance inspection (see 4.6).

d. Periodic group C inspection (see 4.7)

1/ Polarity symbol shall be near the positive terminal and may be placed on either end of the last line of type, depending on the marking orientation with respect to the polarity.

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4.3 Inspection conditions.

4.3.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.3.2 AC measurements. AC measurements shall be made at the frequency specified. The magnitude of the ac voltage shall be equal to or less than 1.0 volt root mean square (rms). The maximum dc bias voltage shall be equal to or less than 2.2 volts for all ac measurements of polarized capacitors.

4.3.3 Reference measurements. When requirements are based on comparative measurements made before and after conditioning, the reference measurement shall be considered the last measurement made at 25°C ±5°C prior to conditioning. Unless reference measurements have been made within 30 days prior to the beginning of conditioning, they shall be repeated.

4.3.4 Power supply. The power supply used for life testing shall have a regulation of ±2 percent or less of the rated voltage. The power source employed for dc leakage current measurements shall be stabilized to at least ±100 parts per million. During measurements there must be no voltage fluctuations of sufficient amplitude to produce a variation in the current measurement as read with any dc leakage current tester used to test capacitors.

4.4 Qualification inspection. Qualification inspection shall he performed at a laboratory acceptable to the Government (see 6.4) on sample units produced with equipment and procedures normally used in production.

4.4.1 Sample size. The number and style combinations of capacitors to be subjected to qualification inspection shall be as specified in the appendix to this specification.

4.4.2 Inspection routine. The sample units shall be subjected to the inspections specified in table III, in the order shown.

4.4.3 Failures. Failures in excess of those allowed in table III shall be cause for refusal to grant qualification approval

4.5 Retention of qualification. To retain qualification the contractor shall forward a report at 6-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery (groups A and B), indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.

b. A summary of the results of tests performed for periodic inspection (group C), including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 6-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

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4.6 Conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A, B, and C inspections except as specified in 4.6.1.4.3, delivery of products which have passed groups A and B inspections shall not be delayed pending the results of group C inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of all capacitors of the same style (see 3.1), produced under essentially the same conditions with the same basic materials, and offered for inspection during a single month. The capacitance values and voltages produced shall be represented in the lot in approximately the ratio of production. Voltage groups shall be as follows:

I - - - - - - - - - - - - - - - - - - - 6 through 20 volts.
II - - - - - - - - - - - - - - - - - - - 25 through 50 volts.

TABLE III. Qualification inspection.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures allowed
Group I	3.5	4.8.2		
Voltage aging				
Construction analysis	3.6	4.8.3	154	
(metal cased units only)			or	0
	3.7	4.8.4	160 1/	
DC leakage	3.8	4.8.5		
Capacitance	3.9	4.8.6		
Dissipation factor	3.1, 3.3, 3.3.1,			
Visual and mechanical inspection	3.4, 3.4.1, 3.4.2, 3.21 and 3.22	4.8.1		
Group II				
Shock (specified pulse)	3.10	4.8.7		
(polarized capacitors only)	3.11	4.8.8	12	
Vibration, high frequency	3.12	4.8.9		
Thermal shock				
Group III				
Resistance to soldering heat	3.13	4.8.10		
Terminal strength	3.14	4.8.11	18	1
Moisture resistance	3.15	4.8.12		
Group IV				
Stability at low and high temperatures	3.16	4.8.13	12	
Surge voltage	3.17	4.8.14		
Group V				
Life (1,000 hours at +85° C)	3.18	4.8.15	102	1
Group VI				
Solderability	3.19	4.8.16		
Resistance to solvents	3.20	4.8.17	10	1

1/ For metal cased units.

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4.6.1.2 Prescreen inspection. All capacitors submitted for group A inspection shall be 100 percent prescreened for the inspections specified in table IV, in the order shown.

TABLE IV. Prescreen inspection.

Inspection	Requirement paragraph	Test method paragraph
Voltage aging	3.5	4.8.2
DC leakage	3.7	4.8.4
Capacitance	3.8	4.8.5
Dissipation factor	3.9	4.8.6

4.6.1.3 Group A inspection. Group A inspection shall consist of the inspections specified in table V, in the order shown.

TABLE V. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure
Subgroup 1			
DC leakage	3.7	4.8.4	
Capacitance	3.8	4.8.5	20 Samples
Dissipation factor	3.9	4.8.6	0 failures
Subgroup 2			
Visual and mechanical inspection	3.1, 3.3, 3.3.1, 3.4, 3.4.1, 3.4.2, 3.21 and 3.22	4.8.3	13 Samples 0 failures
Construction analysis (metal cased units only)	3.6		
Subgroup 3			
Solderability	3.19	4.8.16	13 Samples 0 failures

4.6.1.3.1 Subgroups 1 and 2.

4.6.1.3.1.1 Sampling plan. The sampling plan for subgroups 1 and 2 shall be as specified in table V.

4.6.1.3.1.2 Rejected lots. The rejected lot shall be segregated from new lots and those lots which have passed inspection. The rejected lot shall be 100 percent inspected for those quality characteristics found defective in the sample and any defective found shall be removed from the lot. A new sample of parts shall then be randomly selected in accordance with table V. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

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4.6.1.3.2 Subgroup 3 (solderability).

4.6.1.3.2.1 Sampling plan. Thirteen samples shall be selected randomly from every inspection lot and subjected to the subgroup 3 solderability test. The manufacturer may use electrical rejects from the subgroup 1 screening test for all or part of the samples to be used for solderability testing. If there are one or more defects, the lot shall be considered to have failed.

4.6.1.3.2.2 Rejected lots. In the event of one or more defects, the inspection lot shall be rejected. The manufacturer may use one of the following options to reword the lot:

a. Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.6.1.3.2.1. Production lots that pass the solderability test are available for shipment. Production lots failing the solderability test can be reworked only if submitted to the solder dip procedure in item b.

b. The manufacturer shall submit the failed lot to a 100 percent solder dip using an approved solder dip process in accordance with 3.4.2.1. Following the solder dip, the electrical measurements required in group A, subgroup 1 tests shall be repeated on another 125 samples with no defects allowed. Thirteen additional samples shall then be selected and subjects to the solderability test with no defects allowed. If the lot fails this solderability test, the lot shall be considered rejected and shall not be furnished against the requirements of this specification.

4.6.1.3.2.3 Disposition of samples. The solderability test is considered a destructive test and samples submitted to the solderability test shall not be supplied on the contract.

4.6.1.4 Group B inspection. Group B inspection shall consist of the inspections specified in table VI in the order shown, and shall be made on sample units which have been subjected to and have passed the group A inspection.

TABLE VI. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure
Subgroup 1 Stability at low and high temperatures	3.16	4.8.13	8 samples 0 failures
Subgroup 2 Surge voltage	3.17	4.8.14	8 samples 0 failures

4.6.1.4.1 Sampling plan. The sampling plan shall be as specified in table VI.

4.6.1.4.2 Rejected lots. The rejected lot shall be segregated from new lots which have passed inspection. The contractor may rework it to correct the defects, or screen out the defective units, and resubmit for inspection in accordance with table VI. A new sample of parts shall then be randomly selected in accordance with table VI. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.4.3 Disposition of sample units. Sample units which have been subjected to subgroup 2, group B inspection shall not be delivered on the contract.

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TABLE VII. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph	Units to be inspected	No. of defectives allowed
Subgroup 1				
Shock (specified pulse) (polarized capacitors only) 1/	3.10	4.8.7	12	1
Vibration, high frequency 1/	3.11	4.8.8		
Thermal shock 1/	3.12	4.8.9		
Subgroup 2				
Resistance to soldering heat 2/	3.13	4.8.10	18	
Terminal strength 2/	3.14	4.8.11		
Moisture resistance 2/	3.15	4.8.12		
Subgroup 3				
Resistance to solvents 1/	3.20	4.8.17	10	
Subgroup 4				
Life (1,000 hours at +85°C) 1/	3.18	4.8.15	25	1

1/ If the manufacturer can demonstrate that this test has been performed five consecutive times with zero failures, the frequency of this test, with the approval of the qualifying activity, can be performed on an annual basis. If the design, material, construction, or processing of the part is changed, or if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency.

2/ If the manufacturer can demonstrate that this test has been performed five consecutive times with zero failures, this, with the approval of the qualifying activity, can be deleted. The manufacturer, however, shall perform this test every three years after the deletion as part of long term design verification. If the design, material, construction or processing of the part is changed or, if there are any quality problems, the qualifying activity may require resumption of the specified testing. Deletion of testing does not relieve the manufacturer from meeting the test requirement in case of dispute.

4.7 Periodic group C inspection. Group C inspection shall consist of the tests specified in table VII, in the order shown. Group C inspection shall be made on sample units selected from lots which have passed group A and group B inspection. Group C samples shall be representative of production. The allowable number of defectives shall be as indicated in table VII.

4.7.1 Sampling plan. Sixty-seven sample units of each style shall be selected from the first lot produced, and thereafter from each production of 500,000 units or once every 3 months whichever occurs first. The 67 sample units shall be subdivided as specified for each subgroup in table VII.

4.7.1.1 Disposition of sample units. Sample units which have been subjected to group C inspection, shall not be delivered on the contract.

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4 7.2 Noncompliance. If a sample fails to pass group C inspection the manufacturer shall immediately notify the qualifying activity and the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspection may be reinstituted, however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.8 Methods of inspection.

4.8.1 Visual and mechanical inspection. Capacitors shall be inspected to verify that the materials, design, construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.4.1, 3.4.2, 3.21, and 3.22).

4.8.2 Voltage aging (see 3.5). Capacitors shall be subjected to a minimum of 125 percent of dc rated voltage for 2 hours minimum, at a temperature of 85°C ±5°C. The voltage aging circuit shall have a series resistance of 3.0 ohms maximum. Power source shall not be less than 5 amperes. Capacitors shall then be stabilized at room temperature and examined for evidence of mechanical damage.

4.8.3 Construction analysis (metal cased units only) (see 3.6). Six pieces from each inspection lot shall be dissected in such a way that the plastic end-fill can be inspected to determine the extent of the filling.

4.8.4 DC leakage (see 3.7). DC leakage shall be measured using the dc rated voltage ±2 percent at the applicable test temperature (see 3.1), after a maximum electrification period of 5 minutes. A 1,000 ohm resistor shall be placed in series with the capacitor to limit the charging current. A steady source of power, such as a regulated power supply, shall be used. Unless otherwise specified (see 3.1), measurement accuracy shall be within ±2 percent or 0.02 microamp (μA) which ever is greater (see 4.3.4).

4.8.5 Capacitance (see 3.8). Capacitors shall be tested in accordance with method 305 of MIL-STD-202. Unless otherwise specified (see 3.1), the following details shall apply.

a. Test frequency: 120 ±5 Hz.

b. Limit of accuracy: Measurement accuracy shall be within ±2 percent of the reading.

c. Magnitude of polarizing voltage: Maximum dc bias shall be 2.2 volts for all ac measurements. The magnitude of the dc voltage shall be limited to 1.0 volt rms.

4.8.6 Dissipation factor (see 3.9). The dissipation factor shall be measured at a frequency of 120 ±5 Hz (unless otherwise specified, see 3.1) by means of a polarized capacitance bridge. The bridge shall provide a dial reading accuracy of 0.1 percent dissipation factor and measuring accuracy of (±2 percent of the measured dissipation factor plus 0.1 percent).

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4.8.7 Shock (specified pulse) (see 3.10). Capacitors shall be tested in accordance with method 213 of MIL-STD-202. The following details and exceptions shall apply:

a. Special mounting means: Capacitors shall be rigidly mounted on a mounting fixture by the body. (Potting compounds may be used to secure the capacitor body as long as the compound does not support the leads.) Leads shall be secured to rigidly supported terminals, spaced so that the length of each lead from the capacitor is approximately .375 inch (9.53 mm) when measured from the edge of the supporting terminal. Leads shall be within 30 degrees of being parallel. When securing leads, care shall be taken to avoid pinching the leads.

b. Test condition letter I (100 g's peak).

c. Measurements and electrical loading during shock: During the test, observations shall be made to determine intermittent contact or arcing or open-circuiting or short-circuiting. Detecting equipment shall be sufficiently sensitive to detect any interruption with a duration of 0.5 ms. The dc rated voltage (see 3.1) shall be applied to the capacitors during the test.

d. Examinations after test: Capacitors shall be visually examined for evidence of arcing, breakdown, and mechanical damage.

4.8.8 Vibration, high frequency (see 3.11). Capacitors shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting of specimens: Capacitors shall be mounted on a fixture by the body. Leads shall be supported by rigidly supported terminals.

b. Electrical load conditions: During the test, the specified dc rated voltage (see 3.1) shall be applied to the capacitors.

c. Test condition letter D (20 g's).

d. Duration and direction of motion: Four hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the axis.

e. Measurements during vibration: During the last cycle, an electrical measurement shall be made to determine intermittent operation or open- or short-circuiting. Observations shall also be made to determine intermittent contact or arcing or open- or short-circuiting. Detecting equipment shall be sufficiently sensitive to detect any interruption with a duration of 0.5 ms or greater.

f. Measurements after vibration: Not applicable.

g. Examination after test: Capacitors shall be visually examined for evidence of mechanical damage.

4.8.9 Thermal shock (see 3.12). Capacitors shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

a. Test condition letter A.

b. Measurements after thermal shock: DC leakage, capacitance, and dissipation factor shall be measured as specified in 4.8.4, 4.8.5, and 4.8.6, respectively.

c. Examination after test: Capacitors shall be visually examined for evidence of harmful corrosion, mechanical damage, and obliteration of marking (wherever applicable).

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4.8.10 Resistance to soldering heat (see 3.13). Capacitors shall be tested in accordance with method 210 of MIL-STD-202. The following details and exceptions shall apply:

a. Test condition letter B: Procedure 1, leads shall be immersed to within .025 inch (0.64 mm) of the end seal or capacitor body.

b. Cooling time prior to final examinations and measurements after test: Not applicable.

c. Examination after test: Capacitors shall be visually examined for evidence of mechanical damage.

4.8.11 Terminal strength (see 3.14). Capacitors shall be tested in accordance with method 211 of MIL-STD-202. The following details shall apply, unless otherwise specified (see 3.1):

a. Test condition letter A: The body of the capacitor shall be secured; 2-pound applied force.

b. Test condition letter C: One pound applied force, three bends.

c. Measurements after test: After the test, capacitors shall be visually examined for loosening of the terminals and permanent damage to the terminals.

4.8.12 Moisture resistance (see 3.15). Capacitors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Special mounting not required.

b. Initial measurements: Capacitance as specified in 4.8.5.

c. Number of cycles: Ten continuous cycles except steps 7a and 7b are not required.

d. Loading voltage: Not applicable.

e. Final measurements: After removal from chamber, capacitors shall be dried for 1 hour at room temperature and, within 2 to 6 hours, dc leakage, capacitance, and dissipation factor shall be measured as specified in 4.8.4, 4.8.5, and 4.8.6, respectively.

f. Examination after test: Capacitors shall be visually examined for evidence of harmful corrosion, mechanical damage, and obliteration of marking (wherever applicable).

4.8.13 Stability at low and high temperatures (see 3.16). DC leakage, capacitance, and dissipation factor shall be measured as specified in 4.8.4, 4.8.5, and 4.8.6, respectively, at each of the temperatures specified in table VIII except that dc leakage measurements at -55°C (step 2) are not required. The capacitors shall be brought to thermal stability at each test temperature. Thermal stability will have been reached when no further change in capacitance is observed between two successive measurements taken at intervals of 15 +2, -0 minutes. Capacitors may be dried at 85°C for 30 +4, -0 minutes prior to start of tests, however, the decision to reject the parts shall be made on those parts that have been dried.

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TABLE VIII. Temperature for stability test.

Step	Test temperature
	(°C)
1 - - - - - - - - -	+25 ±2
2 - - - - - - - - -	-55, +0 -3
3 - - - - - - - - -	+25 ±2
4 - - - - - - - - -	+85, +4 -0
5 - - - - - - - - -	+25 ±2

4.8.14 Surge voltage (see 3.17). Capacitors shall be subjected to 1,000 cycles of the applicable surge voltage specified in table I. The ambient temperature during cycling (see 3.1), shall be +85°C ±5°C. Each cycle shall consist of a 30 +2, -0 second surge voltage application followed by a 30 +2, -0 seconds discharge period. Voltage application shall be made through a resistor of 33 ohms. The tolerance of the resistor shall be ±5 percent. Each surge voltage cycle shall be performed in such a manner that the capacitor is shorted terminal to terminal through a copper bar, or an equivalent low resistance at the end of the 30 +2, -0 seconds application. An alternate method of shorting the capacitors is discharge through the same resistance that is utilized for charging. After the final cycle, the capacitors shall be stabilized at the inspection conditions specified in 4.3, and the dc leakage, capacitance, and dissipation factor shall be measured as specified in 4.8.4, 4.8.5, and 4.8.6, respectively.

4.8.15 Life (see 3.18). Capacitors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Test temperature and tolerance: Capacitors subjected to the test of group V of table III shall be tested at +85°C +4, -0.

b. Operating conditions: DC rated voltage (see 3.1), shall be applied gradually (not to exceed 5 minutes either by a slow buildup of the voltage or through a resistor which shall be shorted out within 5 minutes). Voltage shall be applied continuously, except for measurement periods. The impedance of the voltage source, as seen from the terminals of each capacitor, shall not exceed 3 ohms. Storage batteries or an electronic power supply capable of supplying at least 1 ampere when a capacitor is shorted shall he used.

c. Test condition letter D (1,000 hours).

d. Measurements during exposure: DC leakage shall be measured as specified in 4.8.4 at 0; 240 +48 -0; 1000 +48 -0 hours. The manufacturer may option to measure dc leakage at 85°C; however, if the capacitors fail to meet the dc leakage requirement in 3.18 the measurement shall be made at 25°C.

e. Measurements after exposure: Capacitors shall be returned to the inspection conditions specified in 4.3 and visually examined for evidence of harmful corrosion, mechanical damage, or obliteration of marking (if applicable); dc leakage, capacitance and dissipation factor shall then be measured as specified in 4.8.4, 4.8.5, and 4.8.6, respectively.

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4.8.16 Solderability (see 3.19). Capacitors shall be tested in accordance with method 208 of MIL-STD-202. The following details shall apply:

a. The number of terminations of each capacitor to be tested: 2 (except CX04 shall be 4).

b. Depth of immersion in flux and solder: All terminals shall be immersed to within .125 inch (3.18 mm) of the capacitor body.

4.8.17 Resistance to solvents (see 3.20). Capacitors shall be tested in accordance with method 215 of MIL-STD-202. The following exceptions shall apply:

a. Brushing is required.

b. Measurements and examination after test: DC leakage, capacitance, and dissipation factor shall be measured as specified in 4.8 4, 4 8.5 and 4.8.6, respectively. The marking shall remain legible and shall not smear.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6 NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. Capacitors covered by this specification are intended mainly for use in filter, bypass, coupling, blocking, and other low-voltage applications (such as transistor circuits) where stability, size, weight, and shelf life are important factors. These capacitors are intended to be used only where supplemental moisture protection is provided or for noncritical applications where hermetic moisture protection is not required.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the specification, the applicable specification sheet, and the complete PIN..

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. Packaging requirements (see 5.1).

d. Title, number and date of the applicable specification sheet, and the complete PIN (see 3.1).

6.3 Tin plated finishes. Tin plating is prohibited (see 3.4.2.2) because it may result in tin whisker growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information, see ASTM B545, "Standard Specification for Electrodeposited Coating of Tin."

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6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus (DSCC-VQP), 3990 East Broad Street, Columbus, OH 43216-5000. 1/

6.5 Subject term (key word) listing.

Capacitance
Polarized
Dissipation factor

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

1/ SD-6, Provisions Governing Qualification, is issued for the information of applicants requesting qualification of products. Copies of this publication may be obtained from the Defense Printing Service Detachment Office, Building 4D (NPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.

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APPENDIX

PROCEDURE FOR QUALIFICATION INSPECTION

1. SCOPE

1.1 This appendix details the procedure for submission of samples, with related data, for qualification inspection of capacitors covered by this specification. The procedure for extending qualification of the required sample to other capacitors covered by this specification is also outlined herein. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

2. SUBMISSION

2.1 Sample.

2.1.1 Single-style submission. A sample consisting of 154 (160 for metal-cased units) sample units of the highest capacitance value in each voltage rating in each style for which qualification is sought shall be submitted.

2.1.2 Combined-voltage submission. A sample consisting of sample units of the highest capacitance value in each voltage group in each style for which qualification is sought shall be submitted (see table IX).

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APPENDIX

TABLE IX. Combined voltage submission.

Style	Type designation	Number of units	Rated volts
CX01	CX01D686-	77	6
	CX01J186-	77	20
	CX01K126-	77	25
	CX01N565-	77	50
CX02	CX02D337-	77	6
	CX02J107-	77	20
	CX02K686-	77	25
	CX02N226-	77	50
CX03	CX03D475-	80	6
	CX03H685-	80	15
	CX03H226-	80	20
	CX03K106-	80	25
	CX03N104-	80	50
CX04	CX04J156-	77	20
	CX04M685-	77	35
CX05	CX05D476-	77	5
	CX05J156-	77	20
	CX05K106-	77	25
	CX05N475-	77	50
CX06	CX06A106-	77	2
	CX06F107-	77	10
	CX06H686-	77	15
	CX06M226-	77	35
CX12	CX12D686-	77	6
	CX12J156-	77	20
	CX12K106-	77	25
	CX12N475-	77	50
CX16	CX16A106-	77	2
	CX16F107-	77	10
	CX16H686-	77	15
	CX16M226-	77	35

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army MI
Navy - AS, CG, MC, OS, SH
Air Force - 19, 99
DLA - CC

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5910-1872)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49137B	**2. DOCUMENT DATE (YYMMDD)** 971009

3. DOCUMENT TITLE Capacitors, Fixed, Electrolytic (Solid Electrolyte), Tantalum, Molded, Conformal Coated and Metal Cased With Plastic End-Fill, Nonhermetically Sealed, General Specification For

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Incl Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON 732-532-9139 992-9139
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5803 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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